Kinematics Part I: Definitions and Diagrams

1 Core Definitions

Kinematics is the study of motion without considering its causes. We define the state of motion using three fundamental vectors. In this handout, we use d (and $\vec{d}$) for displacement throughout.

1.1 Position and Displacement

- **Position ($\vec{r}$ or $\vec{x}$):** A vector representing the location of an object relative to an origin O.
- **Displacement ($\Delta\vec{d}$):** Defined as the change in position. It is the straight-line vector from the starting point to the ending point: $\Delta\vec{d} = \vec{r}_f - \vec{r}_i$.

1.2 Velocity ($\vec{v}$)

Velocity is the rate of change of displacement. **Average velocity** is defined as:

$$\vec{v}_{\text{av}} = \frac{\Delta\vec{d}}{\Delta t} \tag{1}$$

Instantaneous velocity describes the motion at an instant, this will be covered after you learn calculus.

$$\vec{v} = \frac{d\vec{x}}{dt} \tag{2}$$

Note: Here dx and dt are differentials from calculus, and they are not the same as the displacement $\vec{d}$.

1.3 Acceleration ($\vec{a}$)

Acceleration is the rate of change of velocity. For **uniform acceleration**, the (constant) acceleration is:

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t} \tag{3}$$

2 The v–t Diagram (Velocity-Time)

The $v - t$ diagram is a fundamental tool because its geometric properties correspond directly to physical quantities.

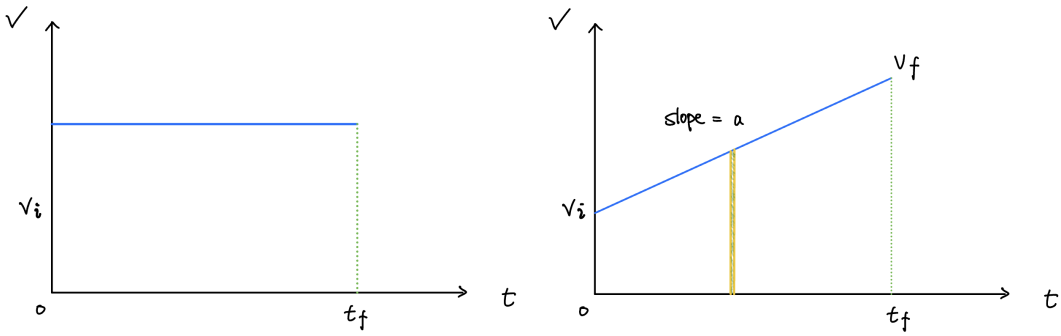

Figure 1: v-t. diagram for uniform motion and uniform acceleration motion

2.1 Why is the Slope equal to Acceleration?

By definition, acceleration is $\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$. In a $v - t$ coordinate system:

- The vertical change is Δv (rise).
- The horizontal change is Δt (run).

Since Slope $= \frac{\text{rise}}{\text{run}} = \frac{\Delta v}{\Delta t}$, the **slope** of a $v - t$ graph represents the **acceleration**.

2.2 Why is the Area equal to Displacement?

Constant Velocity

From Eq. (1), for constant velocity, $\Delta d = v\Delta t$. This geometrically forms a **rectangle** with height v and width Δt. Thus, Area = Displacement (signed area).

Changing Velocity (The "Slicing" Logic)

For an object with changing velocity, we divide the total time into many extremely small intervals dt:

- During each tiny interval, the velocity is nearly constant.
- The displacement for that slice is the area of a very thin rectangle ($v_{\text{slice}} \cdot dt$).
- The total displacement is the sum of all these tiny areas, filling the entire space under the curve.

Conclusion: The **area under the curve** in a $v - t$ diagram represents the **total displacement** $\Delta \vec{d}$ (signed area).

3 Uniform Acceleration Equations

For motion with constant acceleration, we derive the core equations:

3.1 The Velocity Equation

$$v_f = v_i + a\Delta t \tag{4}$$

Derivation: Rearranging the definition of acceleration in Eq. (3): $\Delta v = a\Delta t \Rightarrow v_f - v_i = a\Delta t$.

3.2 The Displacement Equation (Missing v_f)

$$\Delta d = v_i \Delta t + \frac{1}{2}a(\Delta t)^2 \tag{5}$$

Derivation: The total area under the $v - t$ slope is a trapezoid, which can be split into:

1. A **Rectangle**: Area $= v_i \cdot \Delta t$.
2. A **Triangle**: Area $= \frac{1}{2} \cdot \Delta t \cdot \Delta v$.

Substituting $\Delta v = a\Delta t$ from Eq. (4) into the triangle area yields the result.

3.3 The Average Velocity Equation (Missing a)

$$\Delta d = \left(\frac{v_i + v_f}{2}\right)\Delta t \tag{6}$$

Derivation: Geometrically, the area of a trapezoid is $\frac{1}{2}(\text{top} + \text{bottom}) \times \text{height}$.

3.4 The Time-Independent Equation (Missing Δt)

$$v_f^2 = v_i^2 + 2a\Delta d \tag{7}$$

Derivation: Isolate $\Delta t = \frac{v_f - v_i}{a}$ from Eq. (4) and substitute it into Eq. (6).

3.5 The Alternative Displacement Equation (Missing v_i)

$$\Delta d = v_f \Delta t - \frac{1}{2}a(\Delta t)^2 \tag{8}$$

4 The Five-Step Method for Kinematics

To solve motion problems systematically, follow these five steps:

1. Set Positive Direction
Establish a coordinate system (e.g., denote right or up as +). All vectors ($\Delta d, v, a$) must be assigned signs relative to this convention.

2. Draw the $v - t$ Diagram
Sketch the velocity-time graph. This identifies if the motion is multi-stage (e.g., acceleration followed by constant velocity) and helps verify if your final displacement should be positive or negative.

3. Identify Variables (The Inventory)
List the five kinematic variables: $\{v_i, v_f, a, \Delta t, \Delta d\}$. Fill in the given values (with signs!) and mark the target unknown with a "?".

4. Select the Equation
Identify the "missing variable" (the one neither given nor required) and choose the corresponding equation from the table below.

5. Solve and Check
Rearrange the equation algebraically before plugging in numbers. Ensure units are consistent ($m, s, m/s$) and the sign of the result makes physical sense.

Equation Selection Guide (The Big Five)

Equation	**Missing Variable**
$v_f = v_i + a\Delta t$	Δd
$\Delta d = v_i \Delta t + \frac{1}{2}a(\Delta t)^2$	v_f
$v_f^2 = v_i^2 + 2a\Delta d$	Δt
$\Delta d = \left(\frac{v_i + v_f}{2}\right)\Delta t$	a
$\Delta d = v_f \Delta t - \frac{1}{2}a(\Delta t)^2$	v_i